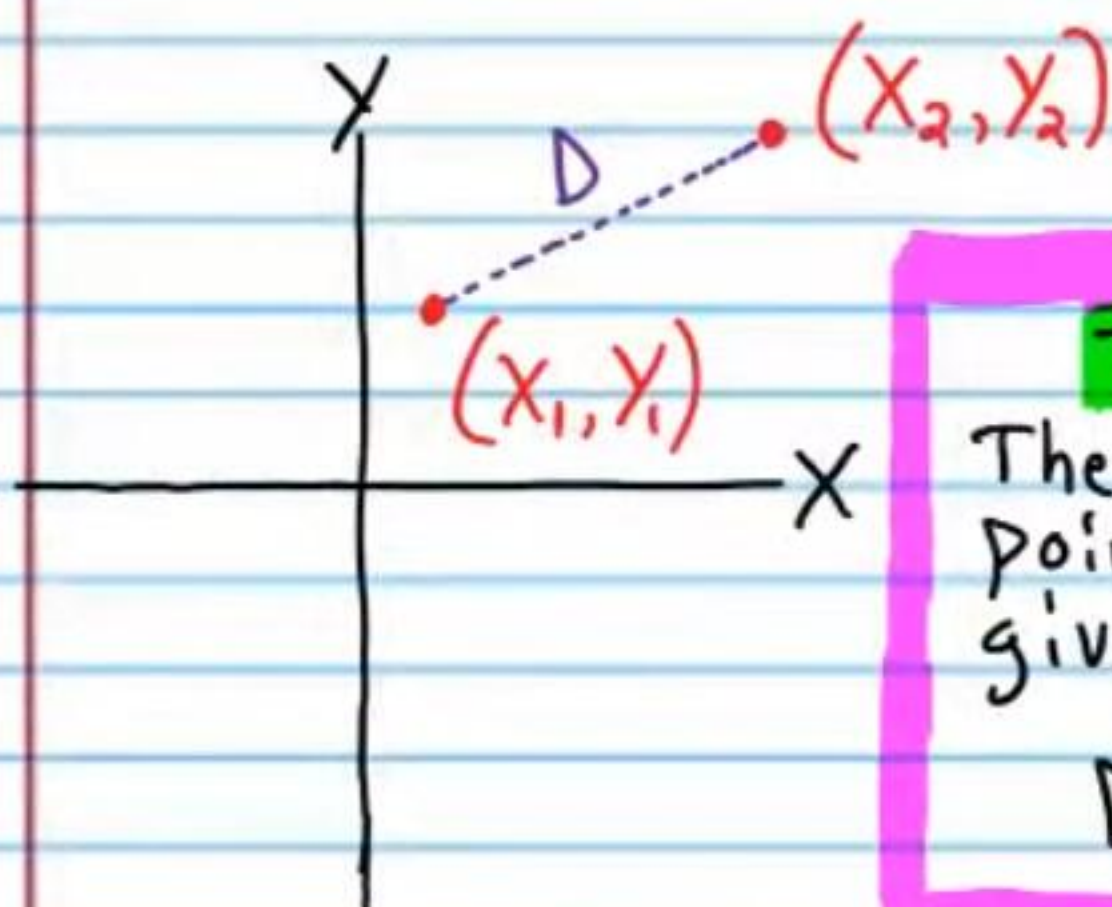


Graphing Quadratic Expressions (Part VI)

Review on the Distance Formula



The Distance Formula

The distance D between two points (x_1, y_1) and (x_2, y_2) is given by the following formula:

$$D = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Find the distance between each pair of points below.

Example I) $(3, -2)$ & $(-4, -1)$

(x_1, y_1) & (x_2, y_2)

$$D = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$D = \sqrt{(-4 - 3)^2 + (-1 - (-2))^2}$$

$$D = \sqrt{(-7)^2 + (-1 + 2)^2}$$

$$D = \sqrt{49 + (1)^2}$$

$$D = \sqrt{49 + 1}$$

$$D = \sqrt{50}$$

$$D = \boxed{5\sqrt{2}}$$

Example II) $(0, 6)$ & $(-5, 8)$

(x_1, y_1) & (x_2, y_2)

$$D = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

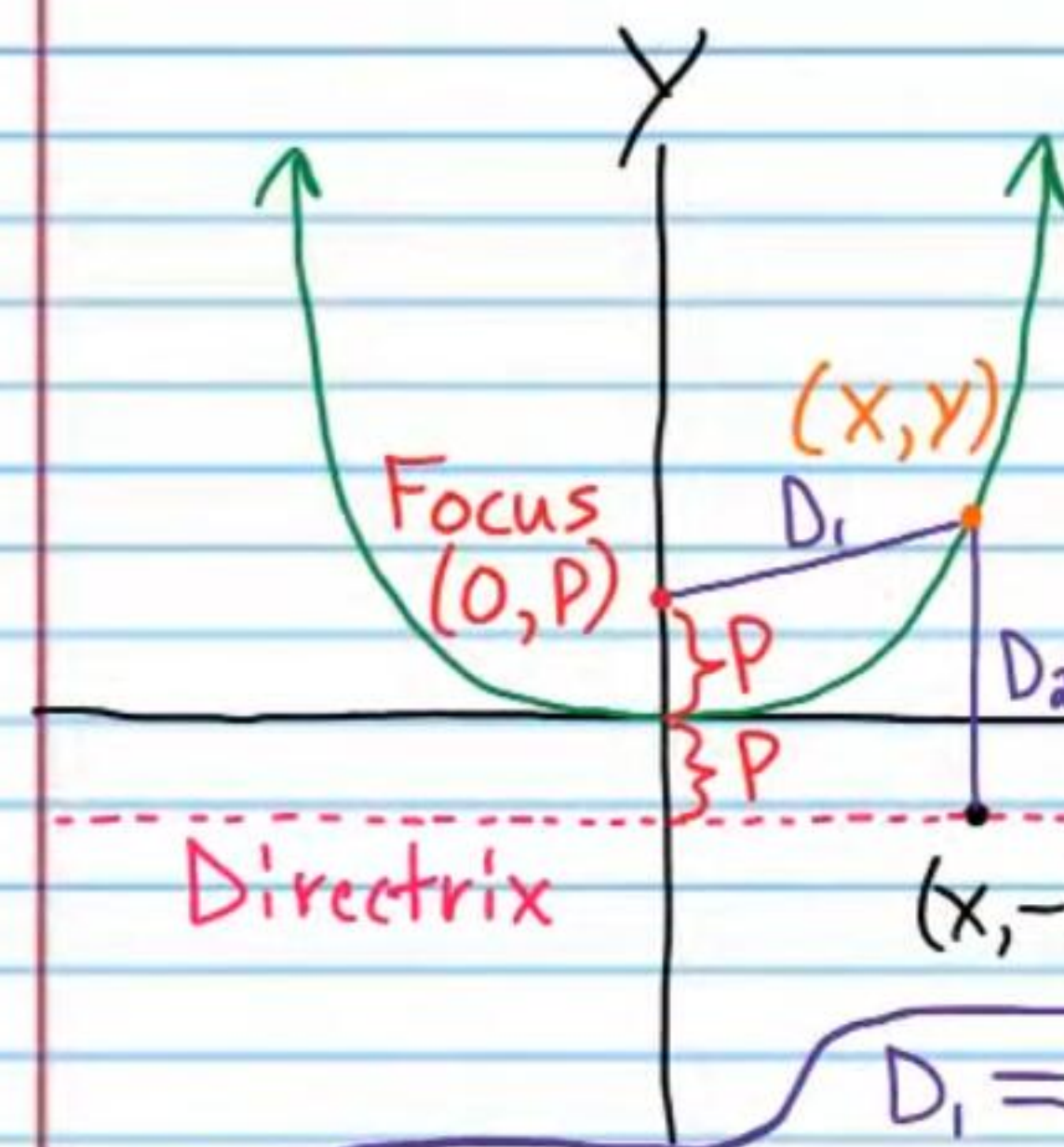
$$D = \sqrt{(-5 - 0)^2 + (8 - 6)^2}$$

$$D = \sqrt{(-5)^2 + (2)^2}$$

$$D = \sqrt{25 + 4}$$

$$D = \boxed{\sqrt{29}}$$

The Exact Definition of a Parabola



Parabola: A parabola is all points (x, y) such that the distance D_1 from a point called the focus to (x, y) is equal to the distance D_2 from (x, y) to a line called the directrix.

$$D_1 = \sqrt{(p - y)^2 + (0 - x)^2}$$

$$D_2 = \sqrt{(y - (-p))^2 + (x - x)^2}$$

$$D_1 = D_2$$

$$\sqrt{(p - y)^2 + (0 - x)^2} = \sqrt{(y - (-p))^2 + (x - x)^2}$$

$$\sqrt{(p - y)^2 + (-x)^2} = \sqrt{(y + p)^2 + (0)^2}$$

$$\sqrt{(p - y)^2 + x^2} = \sqrt{(y + p)^2 + 0}$$

$$\sqrt{(p - y)^2 + x^2} = \sqrt{(y + p)^2}$$

$$(\sqrt{(p - y)^2 + x^2})^2 = (\sqrt{(y + p)^2})^2$$

$$(p - y)^2 + x^2 = (y + p)^2$$

$$p^2 - 2py + y^2 + x^2 = y^2 + 2py + p^2$$

$$-2py + x^2 = 2py$$

$$x^2 = 4py$$

$$\frac{1}{4p} x^2 = y$$

$$\frac{1}{4P} x \cdot x = y$$

$$\frac{1}{4P} (x-0)(x-0) = y$$

$$\frac{1}{4P} (x-m)(x-n) = y$$

$$a(x-m)(x-n) = y$$

Expand

$$\frac{1}{4P} (x-h)^2 + k = y$$

$$a(x-h)^2 + k = y$$

$$\frac{1}{4P} x^2 + bx + c = y$$

$$ax^2 + bx + c = y$$

$$a = \frac{1}{4P}$$

$$\frac{a}{1} = \frac{1}{4P}$$

$$\frac{4P \cdot a}{4P \cdot 1} = \frac{1}{4P}$$

$$\frac{4Pa}{4P} = \frac{1}{4P}$$

$$4Pa = 1$$

$$\frac{4Pa}{4a} = \frac{1}{4a}$$

$$P = \frac{1}{4a}$$

Finding the Focus and Directrix

If a quadratic expression is written in any of the forms $y = ax^2 + bx + c$, $y = a(x-m)(x-n)$, or $y = a(x-h)^2 + k$, and (V_x, V_y) are the coordinates of the vertex, then the coordinates of the focus are $(V_x, V_y + P)$ and the equation of the directrix is $y = V_y - P$ if P is defined to be the expression below.

$$P = \frac{1}{4a}$$

The Three Forms of a Quadratic Expression

Standard Form	Intercept Form	Vertex Form
$y = ax^2 + bx + c$	$y = a(x-m)(x-n)$	$y = a(x-h)^2 + k$

Principles for Graphing Quadratic Expressions

Finding the Vertex in Standard Form

If a quadratic expression is written in the form $y = ax^2 + bx + c$, then the x-value of the vertex is $-\frac{b}{2a}$.

Finding the Vertex in Intercept Form

If a quadratic expression is written in the form $y = a(x-m)(x-n)$, where m and n are real numbers, then the x -value of the vertex is $\frac{m+n}{2}$.

* Intercept form is not commonly used to graph a quadratic expression if the corresponding graph does not move through the x -axis.

Finding the Vertex in Vertex Form

If a quadratic expression is written in the form $y = a(x-h)^2 + k$, then (h, k) is the vertex.

Finding the Y-intercept in Standard Form

If $y = ax^2 + bx + c$ is graphed on an xy -plane, then $(0, c)$ is the y -intercept of the graph.

Graph the following quadratic expressions. Plot the focus and the vertex and draw the directrix and write its equation. You must plot a few other points to draw the parabola.

① $y = -\frac{1}{8}x^2 - x + 2$

$a = -\frac{1}{8}, b = -1, c = 2$

I) Vertex

$$V_x = -\frac{b}{2a} = -\frac{(-1)}{2(-\frac{1}{8})} = -\frac{1}{\frac{2}{8}} = -4$$

$$V_y = -\frac{1}{8}(-4)^2 - (-4) + 2$$

$$V_y = 4$$

$$V(-4, 4)$$

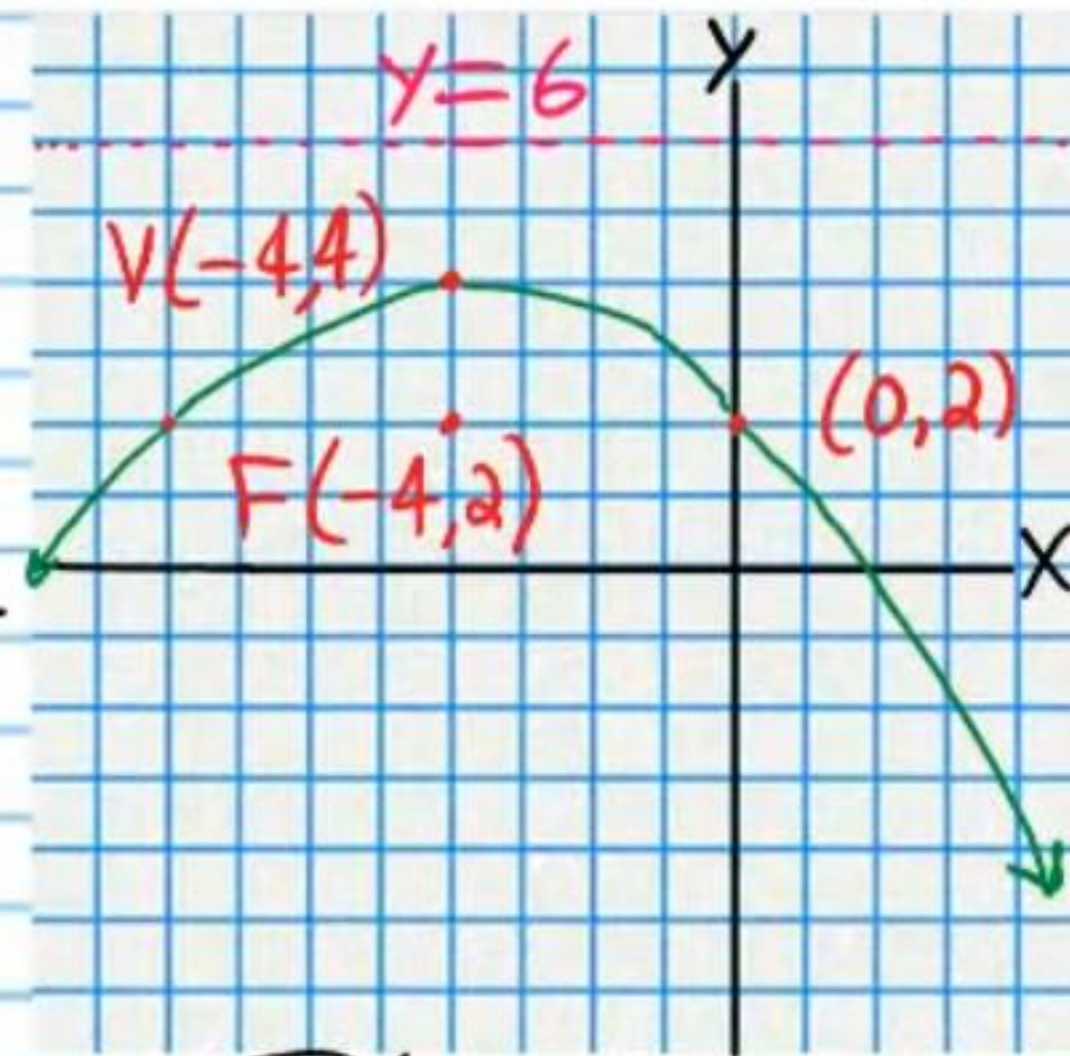
II) $x=0, (0, 2)$

III) Focus

$$p = \frac{1}{4a}$$

$$p = \frac{1}{4(-\frac{1}{8})}$$

$$p = -\frac{1}{4 \cdot \frac{1}{8}} = -2$$



$$p = -2$$

$$F(V_x, V_y + p)$$

$$F(-4, 4 + (-2))$$

$$F(-4, 2)$$

IV) Directrix

$$y = V_y - p$$

$$y = 4 - (-2)$$

$$y = 4 + 2$$

$$y = 6$$

$$\textcircled{2} y = x^2 - 2x - 3$$

$$a = 1, b = -2, c = -3$$

I) Vertex

$$V_x = -\frac{b}{2a} = -\frac{(-2)}{2(1)} = 1$$

$$V_y = (1)^2 - 2(1) - 3$$

$$V_y = -4$$

$$V(1, -4)$$

II) $X = -1$

$$(-1, 0)$$

III) Focus

$$P = \frac{1}{4a}$$

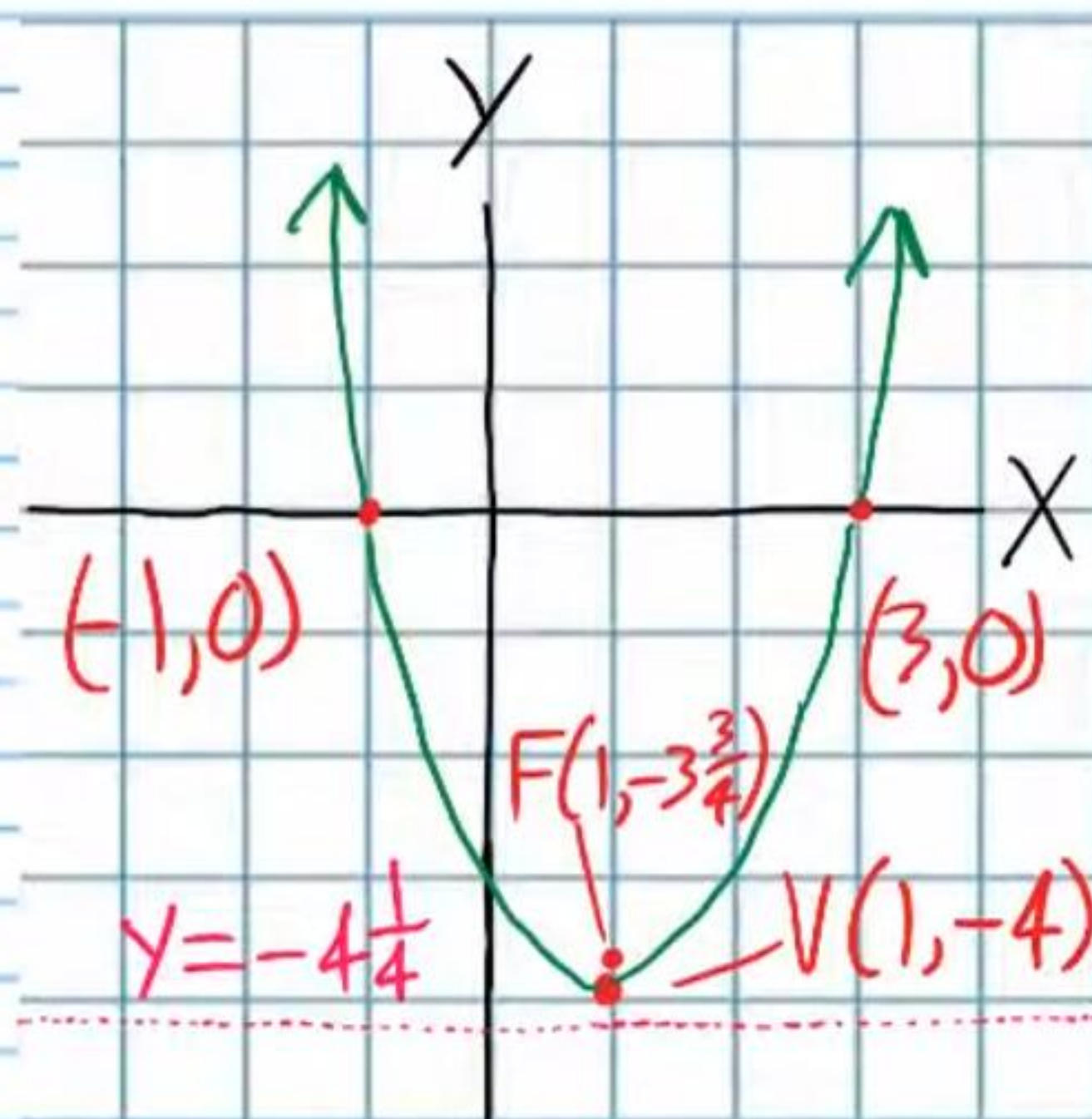
$$P = \frac{1}{4(1)}$$

$$P = \frac{1}{4}$$

$$F(V_x, V_y + P)$$

$$F(1, -4 + \frac{1}{4})$$

$$F(1, -3\frac{3}{4})$$



IV) Directrix

$$y = V_y - P$$

$$y = -4 - \frac{1}{4}$$

$$y = -4\frac{1}{4}$$

$$\textcircled{3} -\frac{1}{4}x^2 - 4x - 13 = y$$

$$a = -\frac{1}{4}, b = -4, c = -13$$

I) Vertex

$$V_x = -\frac{b}{2a} = -\frac{(-4)}{2(-\frac{1}{4})} = \frac{4}{-\frac{2}{4}}$$

$$= -8$$

$$V_y = -\frac{1}{4}(-8)^2 - 4(-8) - 13$$

$$V_y = 3$$

$$V(-8, 3)$$

II) $X = 0, (0, -13)$

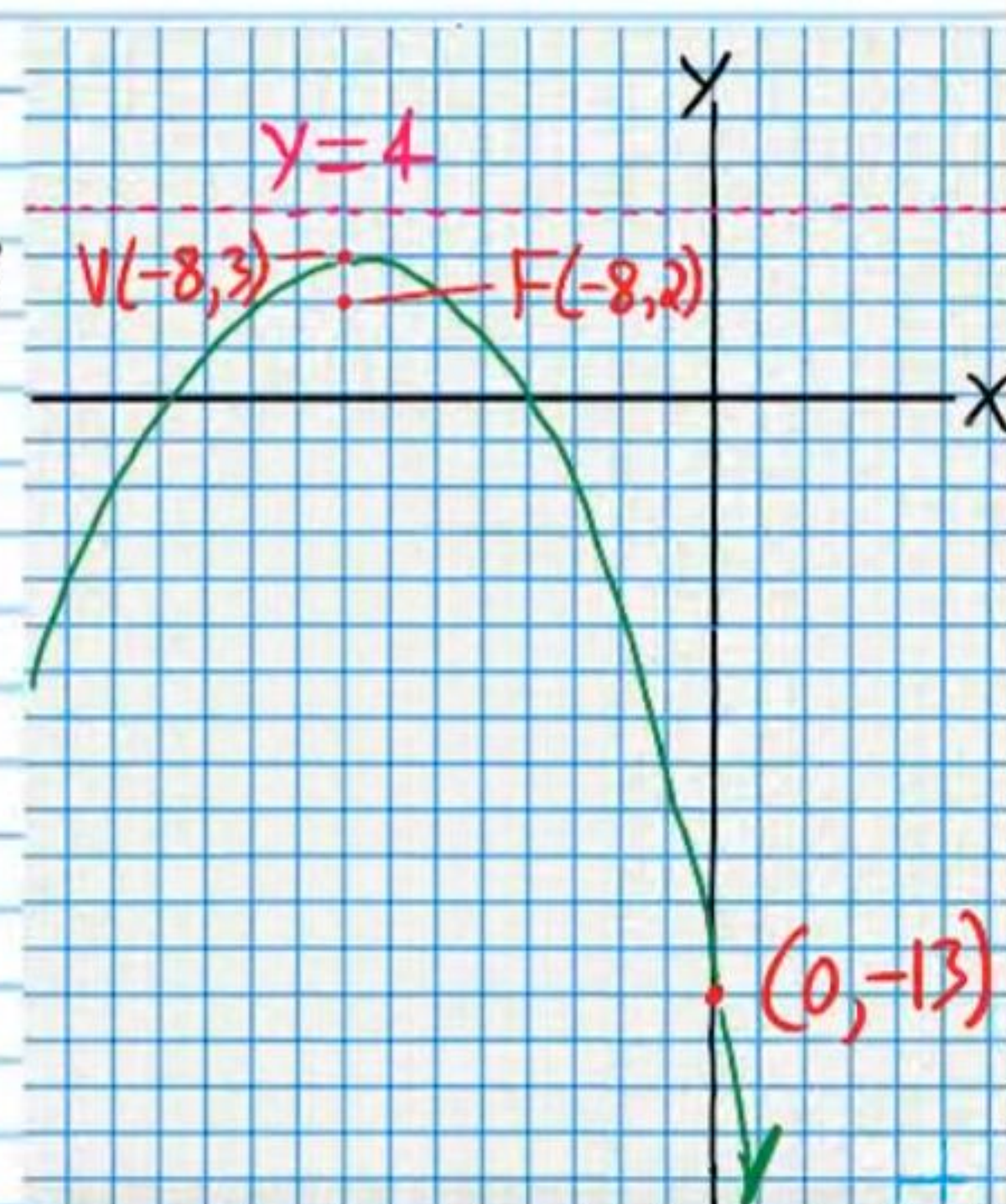
III) Focus

$$P = \frac{1}{4a}$$

$$P = \frac{1}{4(-\frac{1}{4})}$$

$$P = -1$$

$$F(V_x, V_y + P)$$



$$F(-8, 3 + (-1))$$

$$F(-8, 2)$$

IV) Directrix

$$y = V_y - P$$

$$y = 3 - (-1)$$

$$y = 3 + 1$$

$$y = 4$$

$$\textcircled{4} x^2 - 6x + 5 = y$$

$$a=1, b=-6, c=5$$

I) Vertex

$$V_x = -\frac{b}{2a} = -\frac{(-6)}{2(1)} = 3$$

$$V_y = (3)^2 - 6(3) + 5$$

$$V_y = -4$$

$$V(3, -4)$$

II) $x=0, (0, 5)$

III) Focus

$$p = \frac{1}{4a}$$

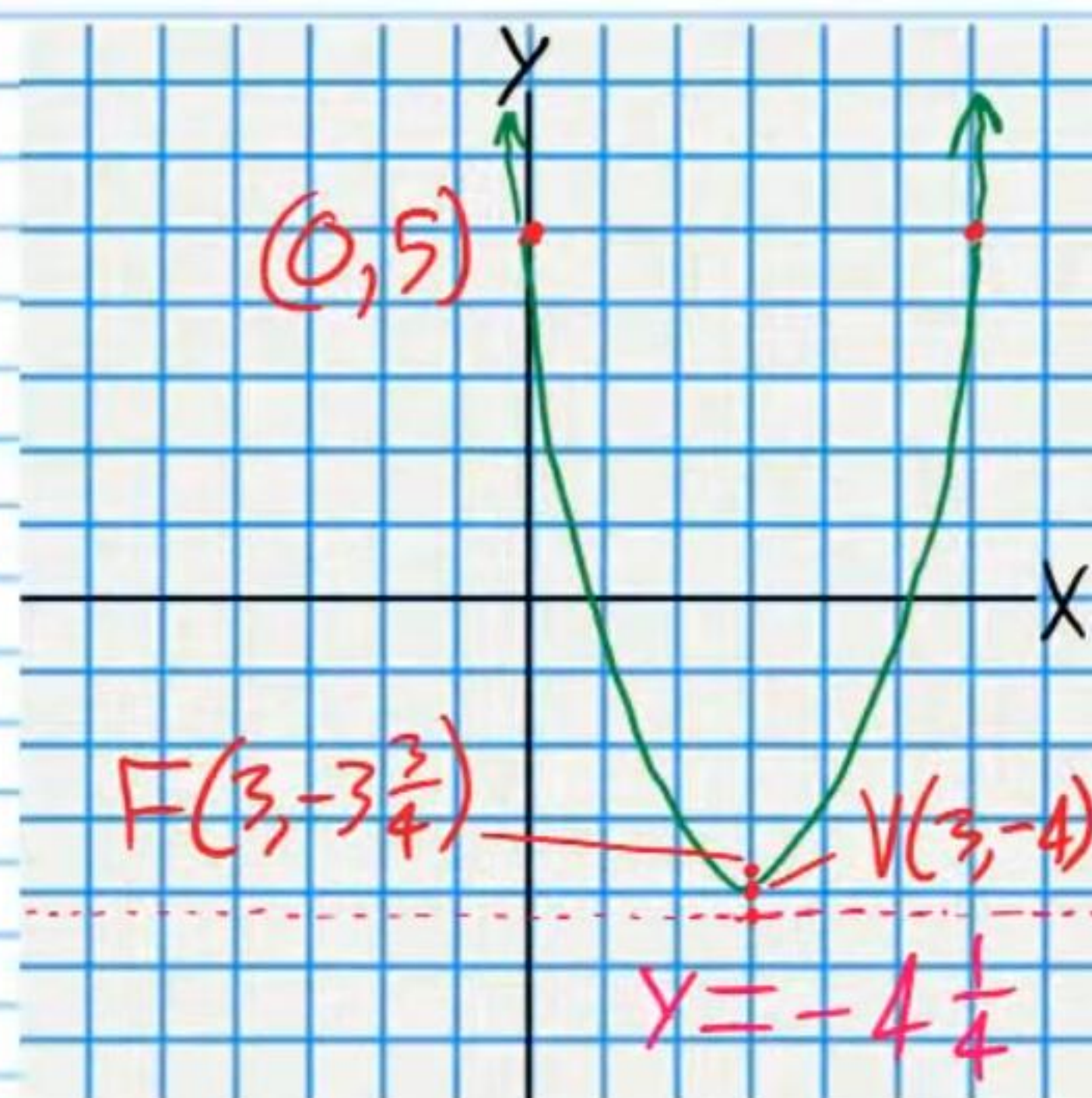
$$p = \frac{1}{4(1)}$$

$$p = \frac{1}{4}$$

$$F(V_x, V_y + p)$$

$$F(3, -4 + \frac{1}{4})$$

$$F(3, -3\frac{3}{4})$$



IV) Directrix

$$y = V_y - p$$

$$y = -4 - \frac{1}{4}$$

$$y = -4\frac{1}{4}$$

$$\textcircled{5} y = 3x^2 + 12x + 10$$

$$a=3, b=12, c=10$$

I) Vertex

$$V_x = -\frac{b}{2a} = -\frac{12}{2(3)} = -2$$

$$V_y = 3(-2)^2 + 12(-2) + 10$$

$$V_y = -2$$

$$V(-2, -2)$$

II) $x=-1, (-1, 1)$

III) Focus

$$p = \frac{1}{4a}$$

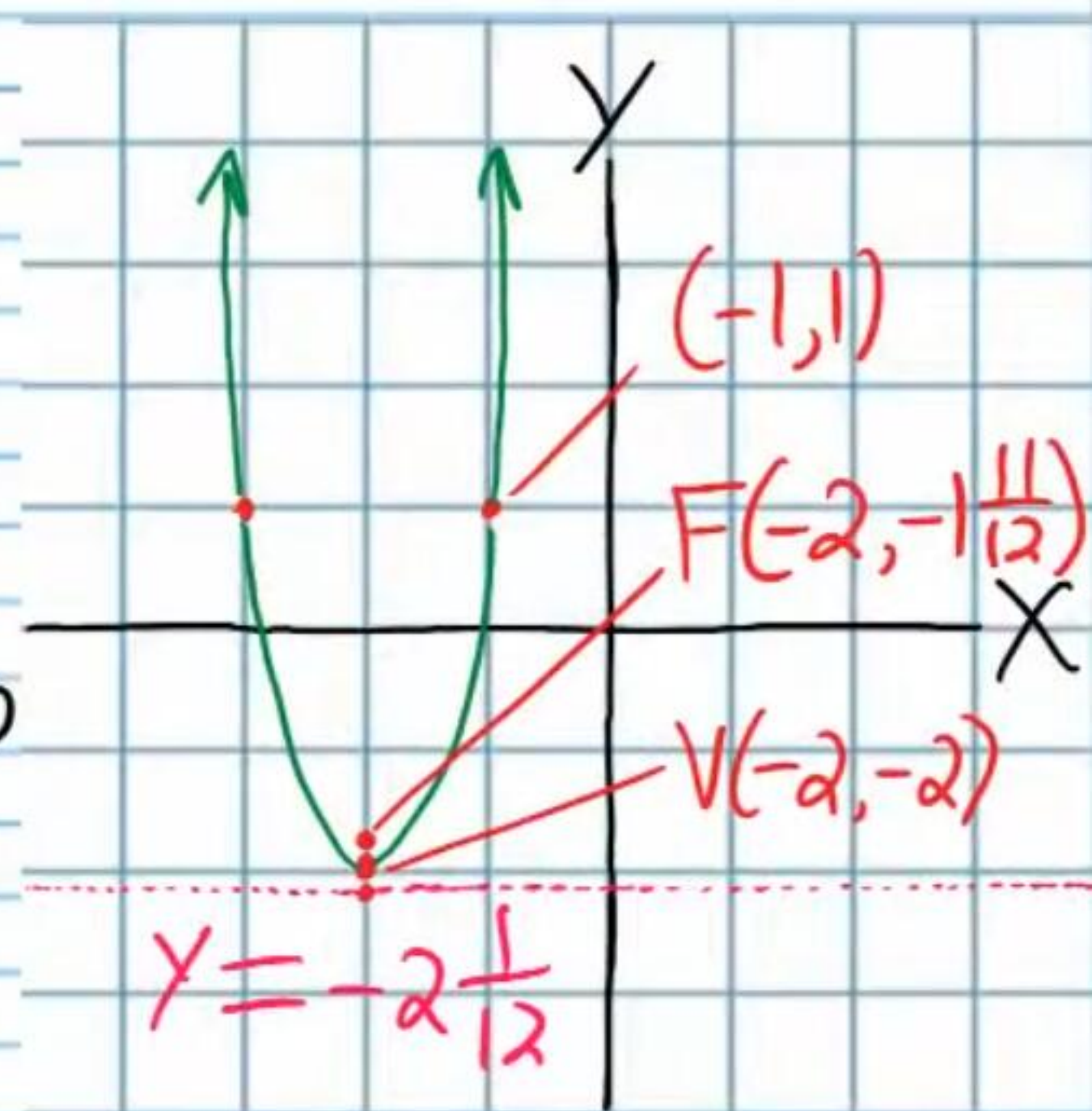
$$p = \frac{1}{4(3)}$$

$$p = \frac{1}{12}$$

$$F(V_x, V_y + p)$$

$$F(-2, -2 + \frac{1}{12})$$

$$F(-2, -1\frac{11}{12})$$



IV) Directrix

$$y = V_y - p$$

$$y = -2 - \frac{1}{12}$$

$$y = -2\frac{1}{12}$$

$$⑥ \quad 4x^2 - 8x + 3 = y$$

$$a = 4, b = -8, c = 3$$

I) Vertex

$$V_x = -\frac{b}{2a} = -\frac{(-8)}{2(4)} = 1$$

$$V_y = 4(1)^2 - 8(1) + 3$$

$$V_y = -1$$

$$V(1, -1)$$

II) $x=0, (0, 3)$

III) Focus

$$P = \frac{1}{4a}$$

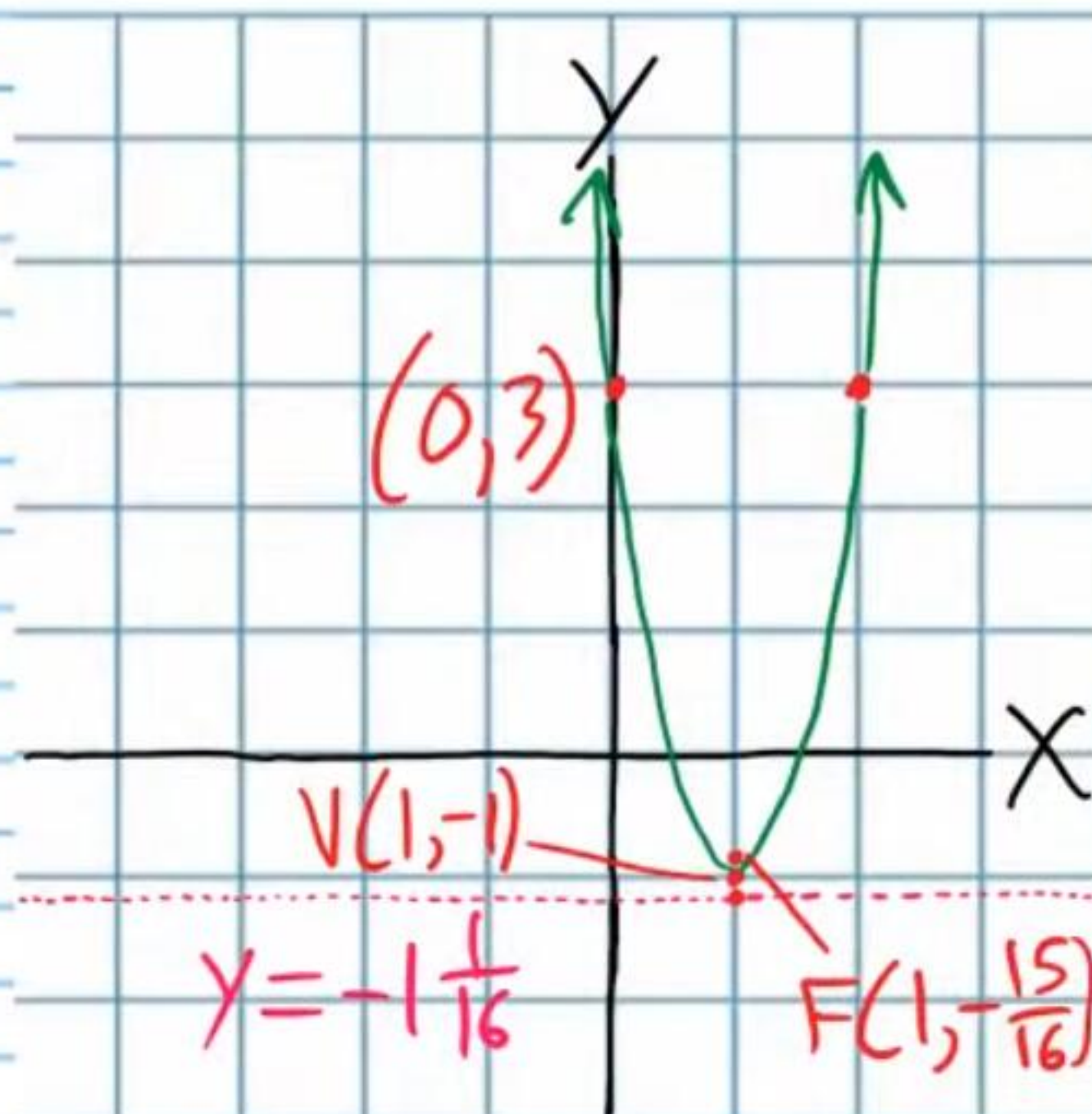
$$P = \frac{1}{4(4)}$$

$$P = \frac{1}{16}$$

$$F(V_x, V_y + P)$$

$$F(1, -1 + \frac{1}{16})$$

$$F(1, -\frac{15}{16})$$



IV) Directrix

$$y = V_y - P$$

$$y = -1 - \frac{1}{16}$$

$$y = -1\frac{1}{16}$$

$$⑦ \quad -\frac{1}{8}(x-2)^2 + 2 = y$$

$$a = -\frac{1}{8}, h = 2, k = 2$$

I) Vertex

$$V_x = h = 2, V_y = k = 2$$

$$V(2, 2)$$

II) $x=6, (6, 0)$

III) Focus

$$P = \frac{1}{4a}$$

$$P = \frac{1}{4(-\frac{1}{8})}$$

$$P = -2$$

$$F(V_x, V_y + P)$$

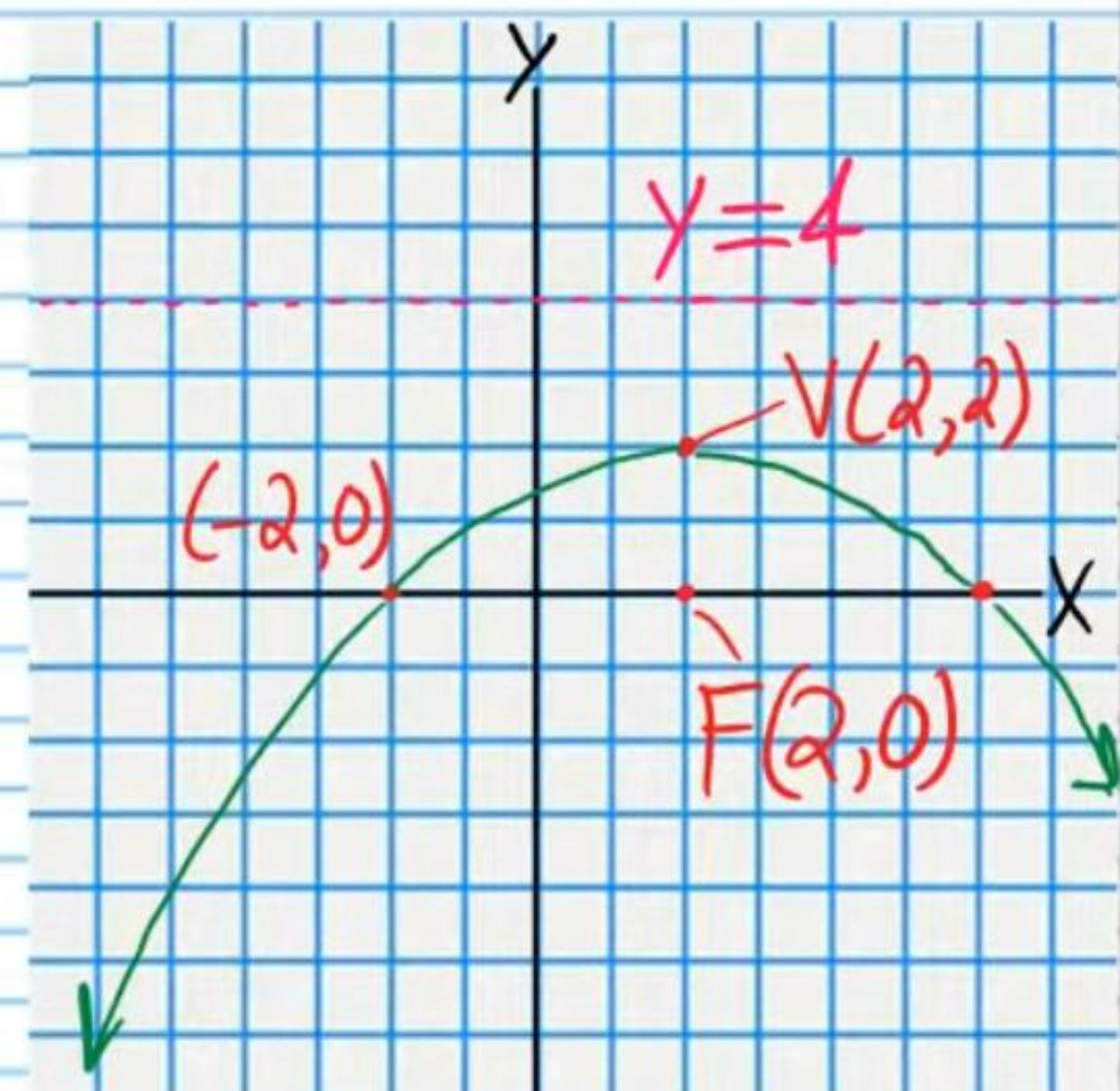
$$F(2, 2 + (-2))$$

$$F(2, 0)$$

IV) Directrix

$$y = V_y - P$$

$$y = 2 - (-2)$$



$$y = 4$$

⑧ $y = 2(x+1)^2 - 3$

$a = 2, h = -1, k = -3$

I) Vertex

$V_x = h = -1, V_y = k = -3$

$V(-1, -3)$

II) $X = 1, (1, 5)$

III) Focus

$P = \frac{1}{4a}$

$P = \frac{1}{4(2)}$

$P = \frac{1}{8}$

$F(V_x, V_y + P)$

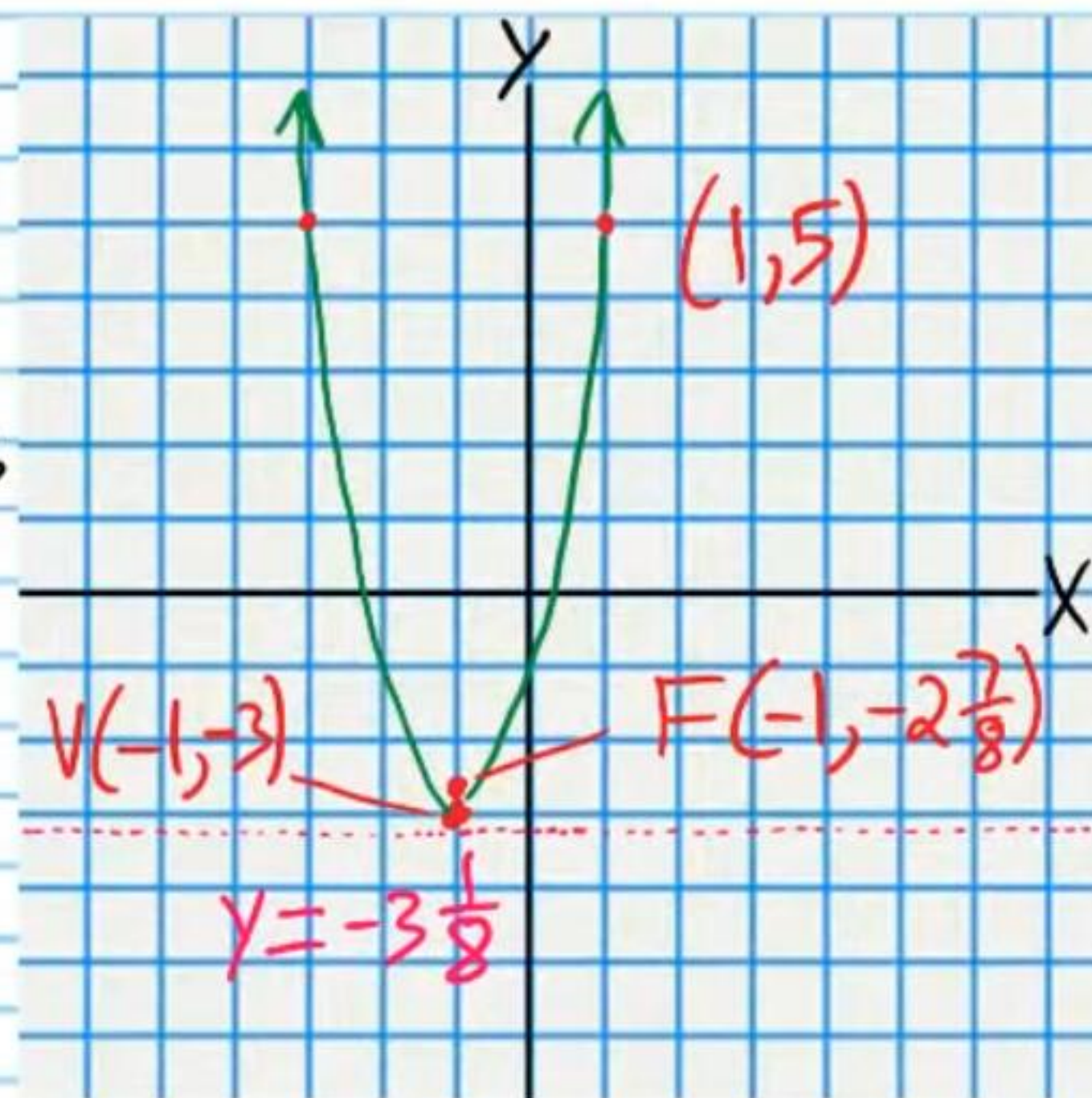
$F(-1, -3 + \frac{1}{8})$

$F(-1, -2\frac{7}{8})$

IV) Directrix

$y = V_y - P$

$y = -3 - \frac{1}{8}$



$y = -3\frac{1}{8}$

⑨ $\frac{1}{2}(x-3)^2 - 2 = y$

$a = \frac{1}{2}, h = 3, k = -2$

I) Vertex

$V_x = h = 3, V_y = k = -2$

$V(3, -2)$

II) $X = 1, (1, 0)$

III) Focus

$P = \frac{1}{4a}$

$P = \frac{1}{4(\frac{1}{2})}$

$P = \frac{1}{2}$

$F(V_x, V_y + P)$

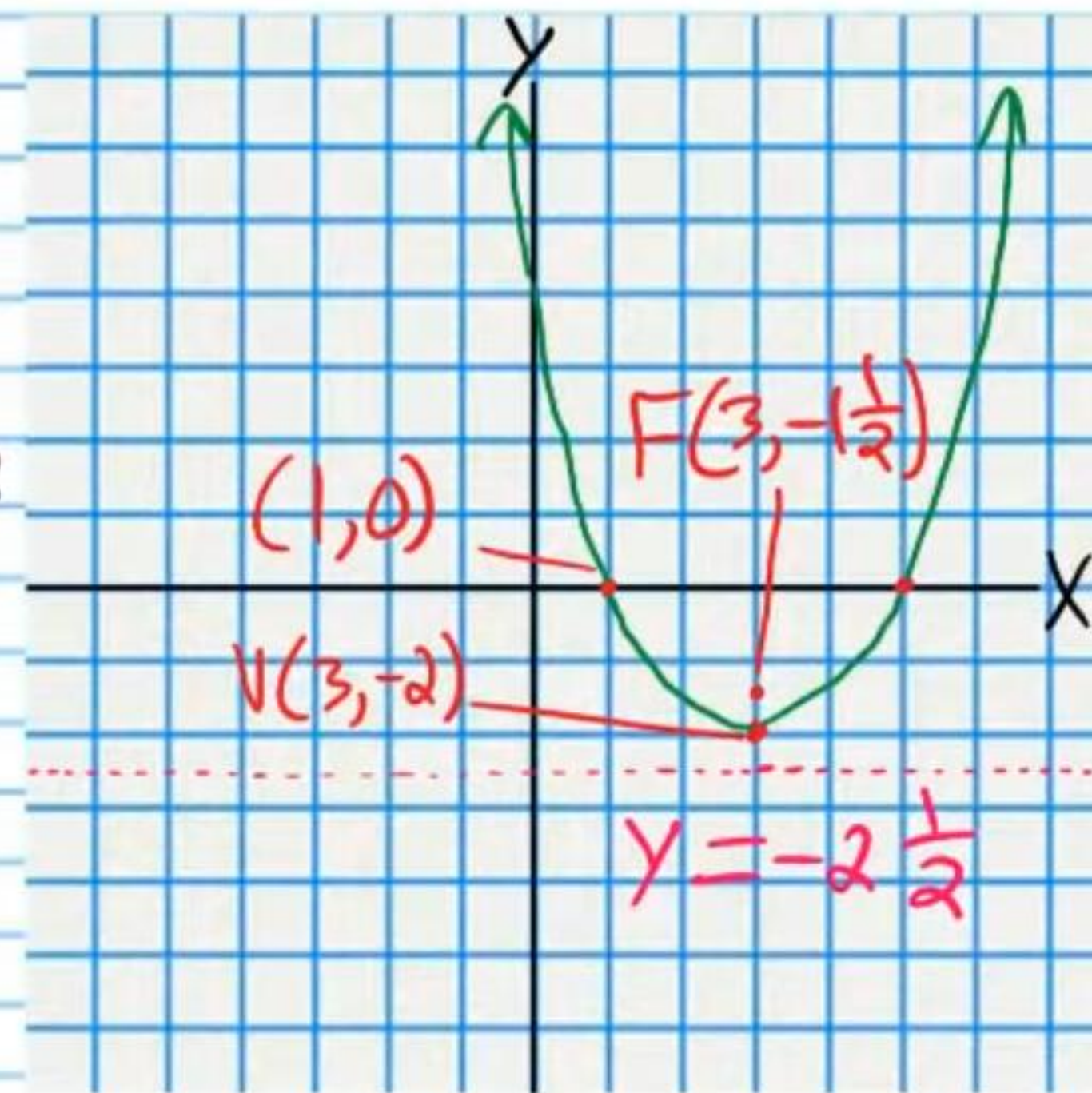
$F(3, -2 + \frac{1}{2})$

$F(3, -1\frac{1}{2})$

IV) Directrix

$y = V_y - P$

$y = -2 - \frac{1}{2}$



$y = -2\frac{1}{2}$

⑩ $y = -(x+1)^2 + 3$

$a = -1, h = -1, k = 3$

I) Vertex

$V_x = h = -1, V_y = k = 3$

$V(-1, 3)$

II) $x = 1, (1, -1)$

III) Focus

$p = \frac{1}{4a}$

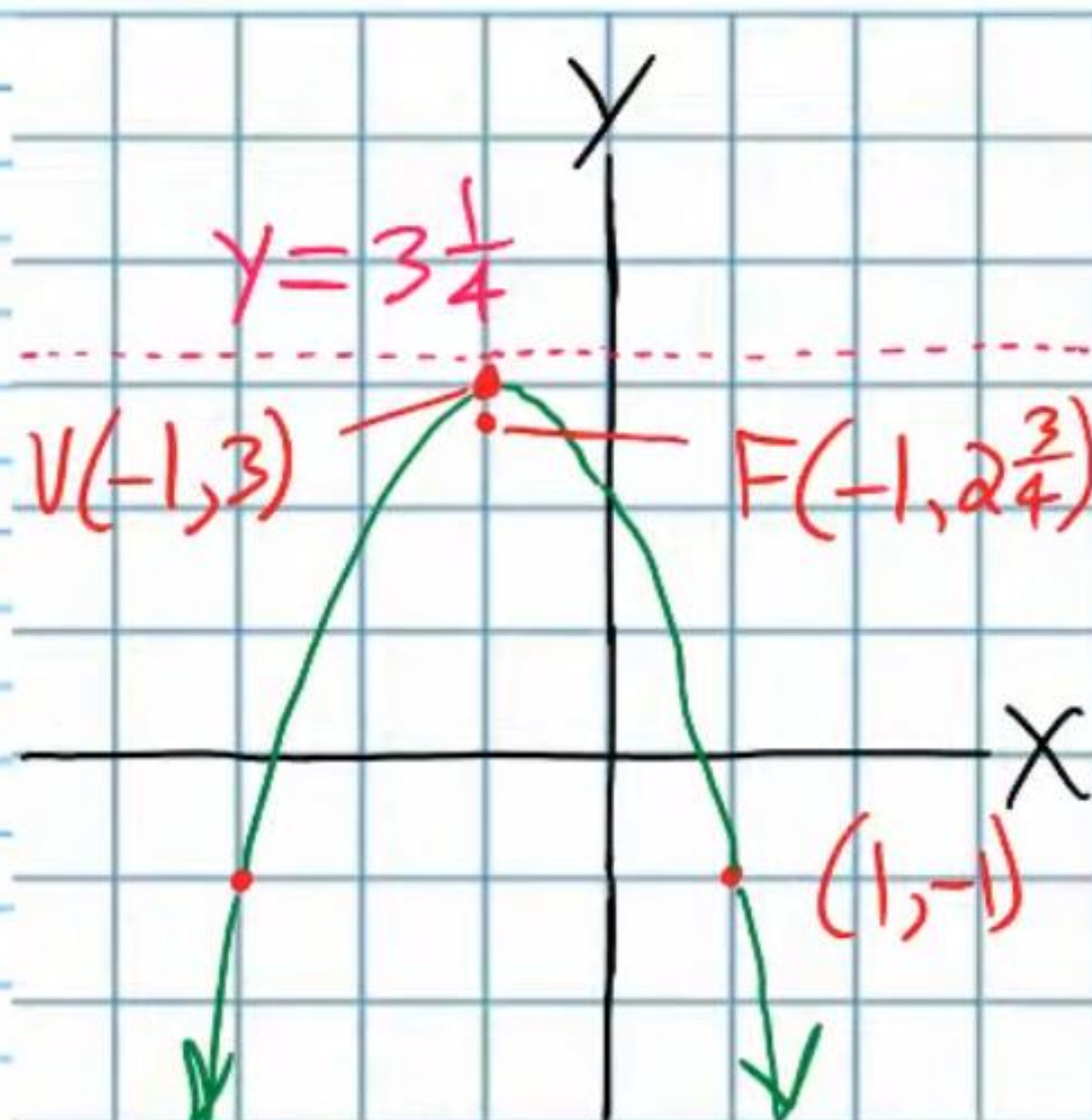
$p = \frac{1}{4(-1)}$

$p = -\frac{1}{4}$

$F(V_x, V_y + p)$

$F(-1, 3 + (-\frac{1}{4}))$

$F(-1, 2\frac{3}{4})$



IV) Directrix

$y = V_y - p$

$y = 3 - (-\frac{1}{4})$

$y = 3\frac{1}{4}$

⑪ $y = (x+1)(x-1)$

$a = 1, m = -1, n = 1$

I) Vertex

$V_x = \frac{m+n}{2} = \frac{-1+1}{2} = 0$

$V_y = (0+1)(0-1)$

$V_y = -1$

$V(0, -1)$

II) $y = 0,$

$(-1, 0) \& (1, 0)$

III) Focus

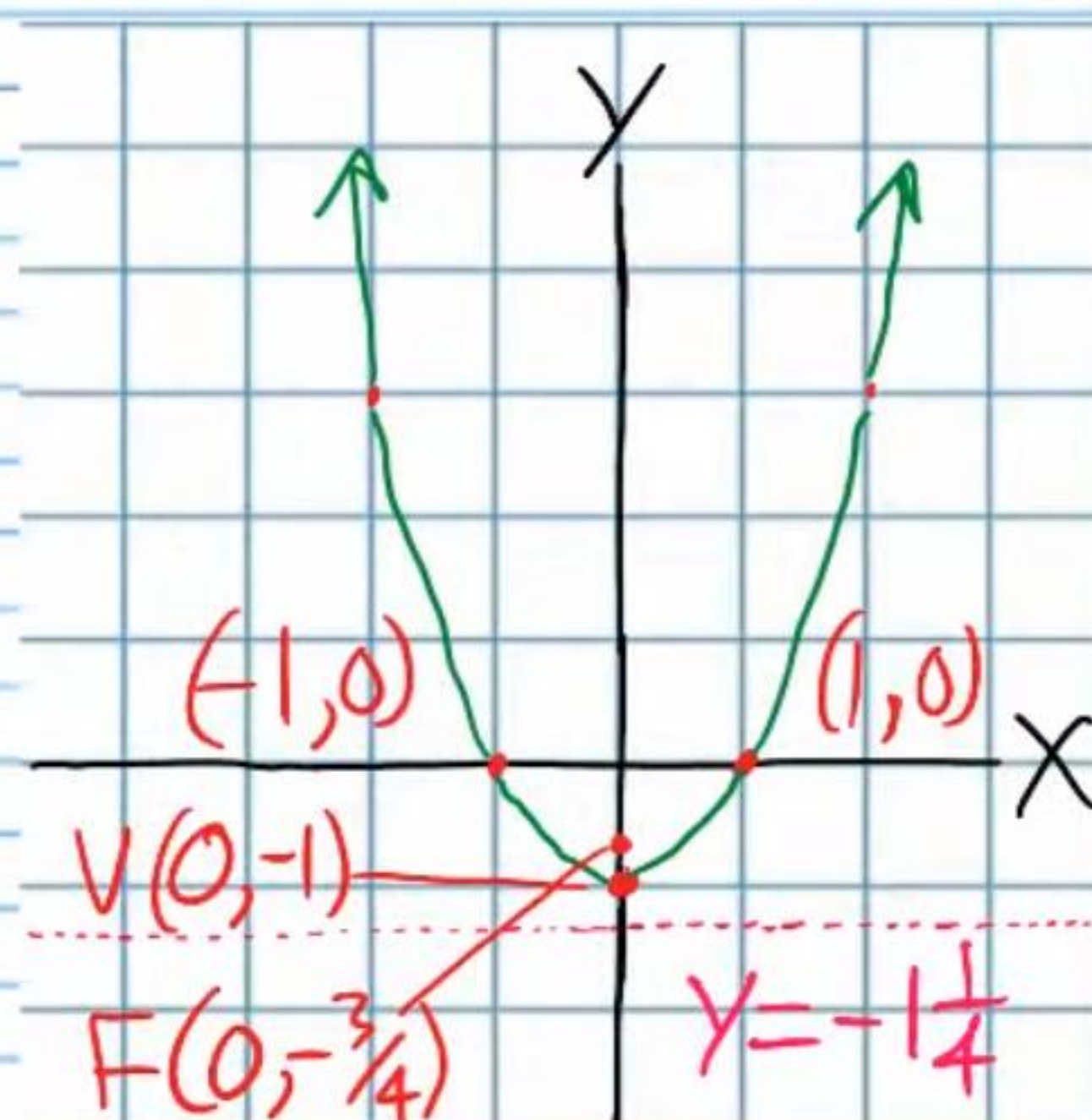
$p = \frac{1}{4a}$

$p = \frac{1}{4(1)}$

$p = \frac{1}{4}$

$F(V_x, V_y + p)$

$F(0, -1 + \frac{1}{4})$



$F(0, -\frac{3}{4})$

IV) Directrix

$y = V_y - p$

$y = -1 - \frac{1}{4}$

$y = -1\frac{1}{4}$

$$\textcircled{12} \frac{1}{4}(x-3)(x-7) = y$$

$$a = \frac{1}{4}, m=3, n=7$$

I) Vertex

$$V_x = \frac{m+n}{2} = \frac{3+7}{2}$$

$$= 5$$

$$V_y = \frac{1}{4}(5-3)(5-7)$$

$$V_y = -1$$

$$V(5, -1)$$

II) $y=0$

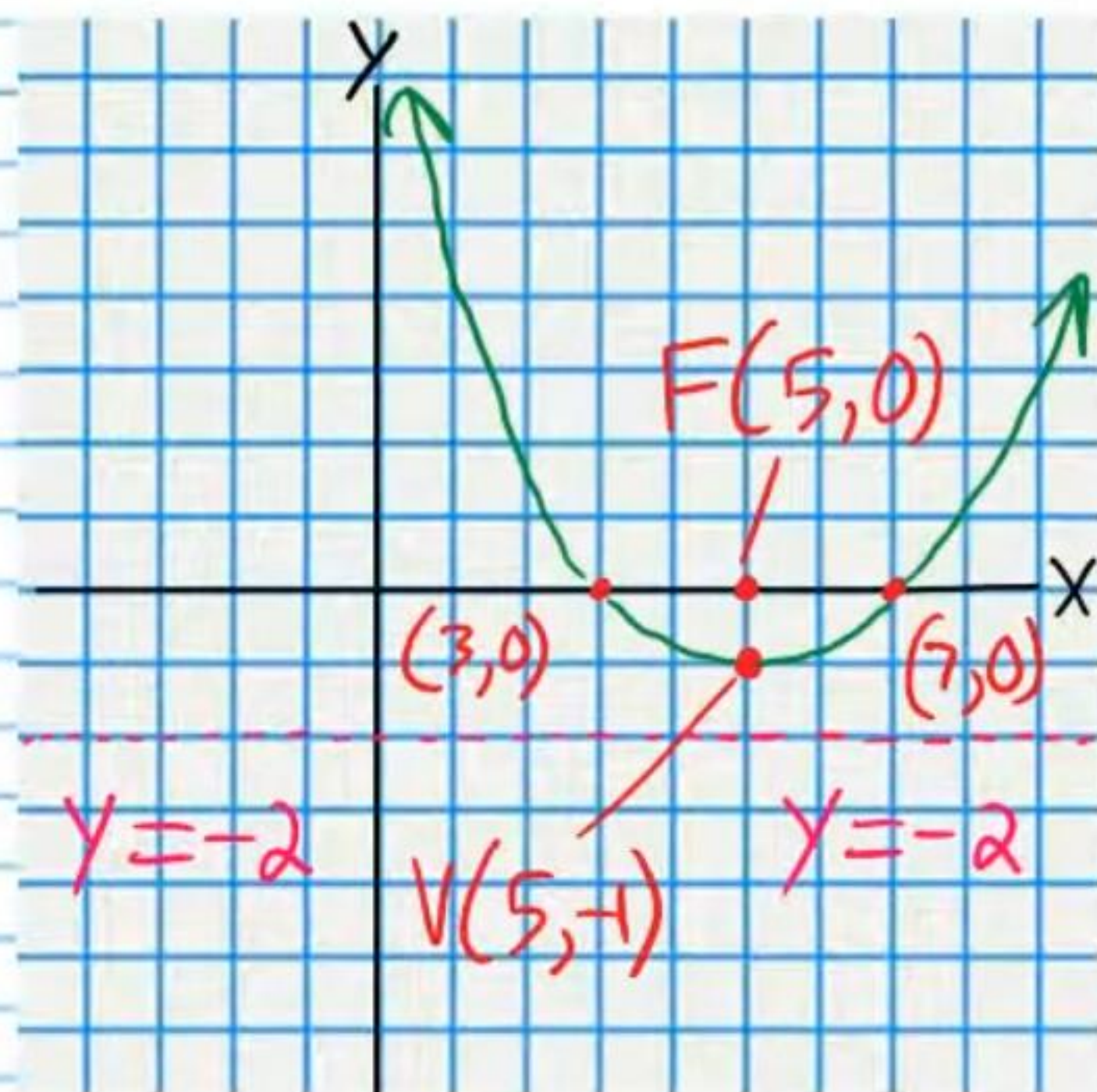
$$(7,0) \text{ \& } (3,0)$$

III) Focus

$$P = \frac{1}{4a}$$

$$P = \frac{1}{4(\frac{1}{4})}$$

$$P = 1$$



$$F(V_x, V_y + P)$$

$$F(5, -1 + 1)$$

$$F(5, 0)$$

IV) Directrix

$$y = V_y - P$$

$$y = -1 - 1$$

$$y = -2$$

$$\textcircled{13} -2(x-3)(x+1) = y$$

$$a = -2, m=3, n=-1$$

I) Vertex

$$V_x = \frac{m+n}{2} = \frac{3+(-1)}{2}$$

$$= 1$$

$$V_y = -2(1-3)(1+1)$$

$$V_y = -2(-2)(2)$$

$$V_y = 8$$

$$V(1, 8)$$

II) $y=0$

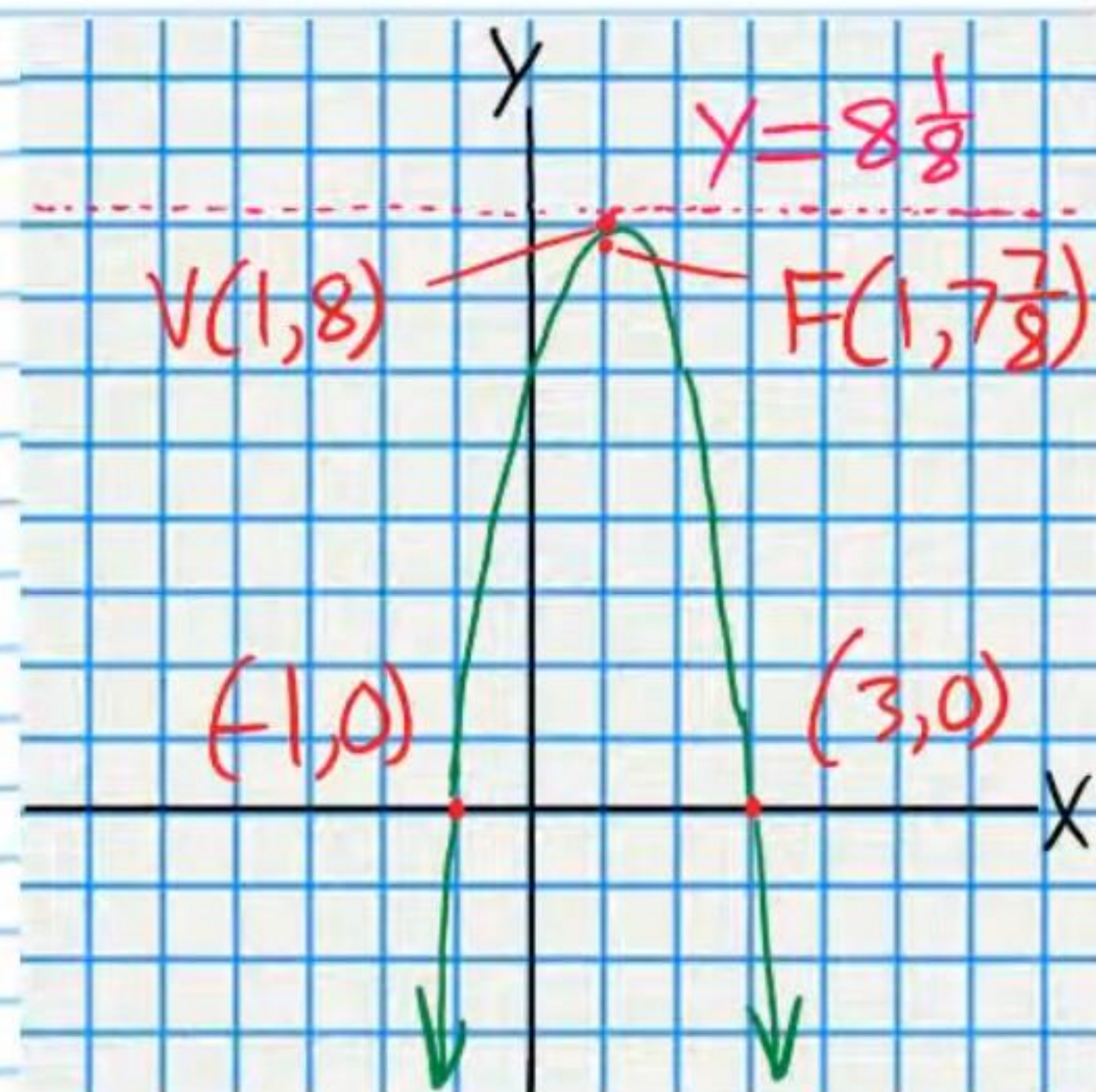
$$(-1,0) \text{ \& } (3,0)$$

III) Focus

$$P = \frac{1}{4a}$$

$$P = \frac{1}{4(-2)}$$

$$P = -\frac{1}{8}$$



$$F(V_x, V_y + P)$$

$$F(1, 8 + (-\frac{1}{8}))$$

$$F(1, 7\frac{7}{8})$$

IV) Directrix

$$y = V_y - P$$

$$y = 8 - (-\frac{1}{8})$$

$$y = 8 + \frac{1}{8}$$

$$y = 8\frac{1}{8}$$

⑭ $y = \frac{1}{3}(x+2)(x+8)$

$a = \frac{1}{3}, m = -2, n = -8$

I) Vertex

$$V_x = \frac{m+n}{2} = \frac{-2+(-8)}{2}$$

$$= -5$$

$$V_y = \frac{1}{3}(-5+2)(-5+8)$$

$$V_y = \frac{1}{3}(-3)(3)$$

$$V_y = -3$$

$$V(-5, -3)$$

II) $y = 0$

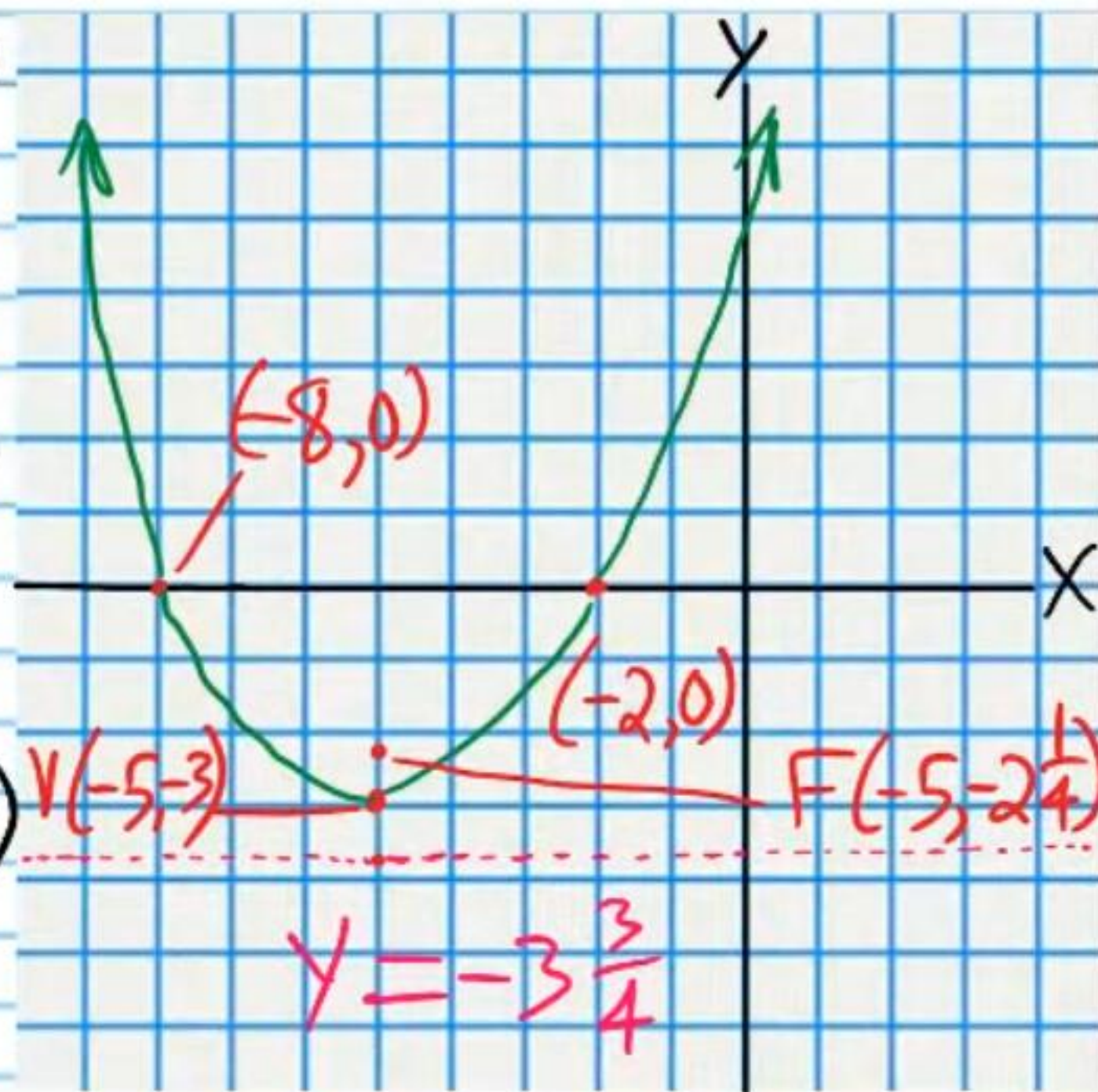
$$(-8, 0) \text{ \& \; } (-2, 0)$$

III) Focus

$$p = \frac{1}{4a}$$

$$p = \frac{1}{4(\frac{1}{3})}$$

$$p = \frac{3}{4}$$



$$F(V_x, V_y + p)$$

$$F(-5, -3 + \frac{3}{4})$$

$$F(-5, -2\frac{1}{4})$$

IV) Directrix

$$y = V_y - p$$

$$y = -3 - \frac{3}{4}$$

$$y = -3\frac{3}{4}$$